

Revenue Forecasting & Business Analysis Summary

Project Overview

The objective of this project was to analyze ERP transaction data, forecast future revenue, and generate business recommendations for production planning and operational decision-making.

The project included:

- Exploratory Data Analysis (EDA)
- Data cleaning and preprocessing
- Feature engineering
- Forecasting model development
- Business interpretation
- Anomaly detection

Data Analysis & Preprocessing

The dataset contained customer information, product details, invoice dates, quantity, price, tax and other charges.

During EDA:

- data types were corrected
- missing values in Tax and Others were handled appropriately
- duplicate records were analyzed carefully because removing them significantly affected total revenue
- negative revenue transactions were identified as returns, cancellations, or invoice reversals
- outliers were detected using the IQR method

Revenue trends and seasonality were analyzed at daily, weekly, and monthly levels. Weekly aggregation showed more stable business patterns compared to highly volatile daily transaction data.

Forecasting Model Development

Initially, Prophet was used for forecasting because it handles seasonality and trend analysis effectively. However, Prophet produced overly smooth forecasts and struggled to capture sudden fluctuations in ERP transaction data.

To improve performance, feature engineering was performed using:

- lag features
- rolling averages
- rolling standard deviation
- calendar-based features such as month, quarter, and weekend indicators

An XGBoost model was then developed using these engineered features. XGBoost performed better because it handled nonlinear relationships and operational volatility more effectively.

Model Results

Prophet Model

- RMSE $\approx$ 14 million
- MAE $\approx$ 10 million

XGBoost Daily Forecast

- RMSE $\approx$ 4.8 Million
- MAE $\approx$ 3.5 Million
- MAPE $\approx$ 20%

XGBoost Weekly Forecast

- MAPE improved to approximately 10.9%
- forecasts became more stable and business-friendly

Weekly forecasting was selected as the final approach because it provided the best balance between forecasting accuracy and operational usefulness.

30-Day Forecast & Anomaly Detection

Using the final XGBoost model, a 30-day revenue forecast was generated to identify future demand patterns and potential business risks.

Statistical anomaly detection using z-scores identified significant revenue spikes. Detailed analysis showed that these spikes were mainly driven by:

- bulk customer orders
- high-demand products
- increased transaction activity

A what-if analysis showed that without these spikes, revenue would have remained close to normal business levels.

Business Recommendations

Based on the analysis, the following recommendations were identified:

- use weekly forecasting for operational planning
- monitor high-risk product lines closely
- track abnormal revenue spikes and dips in real time

- improve inventory and production planning during volatile demand periods
- refresh forecasts regularly using rolling updates

Additional data such as inventory levels, promotions, and supply-chain information could further improve forecasting accuracy.

Conclusion

This project demonstrated how machine learning and time-series forecasting can support business planning using ERP transaction data.

The analysis showed that:

- proper preprocessing and feature engineering are critical for forecasting accuracy
- XGBoost performed better than Prophet for volatile ERP data
- weekly forecasting provided more stable and interpretable business insights
- anomaly detection helped identify important operational events and revenue risks

Overall, the project highlighted the importance of combining data analysis, forecasting models, and business interpretation to support better business decision-making.